

Investigating Robust Overfitting in Adversarial Training

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Definition and Foundation

- Neural networks we have today are extremely powerful, where they are able to achieve high accuracy on complex tasks such as image recognition, yet these modern neural networks happens to also be very vulnerable to attacks, such as adversarial attacks.
- **Adversarial Example:** An input created by adding a small, intentional change to a correctly classified example so that a model gives a wrong answer with high confidence (Goodfellow et al., 1). There will be visual examples later.
- People used to think that neural networks are vulnerable to these attacks because they are *nonlinear*, or they are unpredictable. What this means is that their output does not change in simple proportional steps, so small input changes can sometimes produce disproportionately large or small output changes.
- However, Goodfellow et al. disproved this idea, where they instead found that linearity is actually the problem (especially in high-dimensional input spaces), not because of nonlinear. What linearity means is that an output that changes in direct, proportional steps as input changes, through a simple weighted sum. (Goodfellow et al., 1)
- **How?** Because predictable also means it is easily exploitable. Because a linear model responds to every input in a fixed, proportional way, an attacker can predict in advance which direction to nudge each part of the input to push the output off course. It is similar to how a large crowd cannot tip over a heavy table by each person pushing in a random direction, but the same crowd can tip it over easily if everyone pushes the same way at the same time, even if each individual push is small.

Fast Gradient Sign Method (FGSM)

- **Definition:** A fast, simple method for generating an adversarial example in a single step, rather than through slow, repeated optimization. It works by nudging the input in the exact direction that increases the model's error the most, using only the sign of the gradient.

$$\eta = \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y)), \quad \tilde{x} = x + \eta$$

where, θ : model's trained parameters; x : original input; y : correct label; $J(\theta, x, y)$: cost function used to train model (a measure of how wrong model's prediction is); $\nabla_x J$: gradient of cost function with respect to input (direction that increases error fastest); $\text{sign}(\cdot)$: a function that outputs +1, -1, or 0, indicating direction only, not magnitude; ϵ (**epsilon**): a small constant controlling how large perturbation is allowed to be; η (**eta**): resulting perturbation added to input; $\tilde{x}$ (**x-tilde**): resulting adversarial input.

- Why FGSM is useful: It turned adversarial examples from a slow, expensive computation into something fast enough to use routinely, which made adversarial training practical (Goodfellow et al., 3).

Example 1: Panda to Gibbon (Goodfellow et al., 3)

- Original image: Classified as “panda” with 57.7% confidence.
- A perturbation with magnitude $\epsilon = 0.007$, invisible to the human eye, is added.
- Resulting image: Classified as “gibbon” with 99.3% confidence.
- Two images are visually indistinguishable to human eyes.



x

“panda”

57.7% confidence

+ .007 ×



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

=



$x +$

$\epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

99.3 % confidence

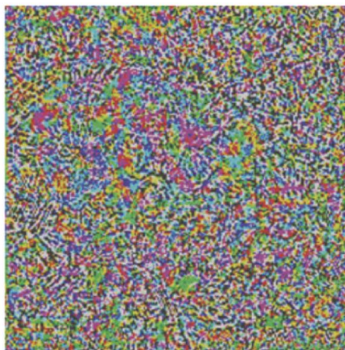
Example 2: Bear to Truck (Veerabadran et al., 2)

- A bear image is correctly classified as "bear." The colored noise is shown 64× larger for visibility. The actual perturbation is nearly invisible.
- After adding the true perturbation, the ANN misclassifies the same image as "truck."
- This reproduces the adversarial example first demonstrated by Goodfellow et al. (2013) and later replicated by Veerabadran et al.
- Veerabadran et al. also found that these perturbations can subtly bias human perception, especially when images are viewed briefly or under uncertainty.

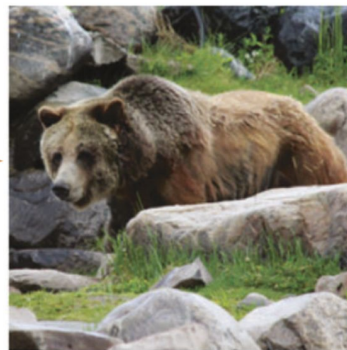
ANN classification: bear



64x



ANN classification: truck



Example 3: MNIST Logistic Regression (Goodfellow et al., 4)

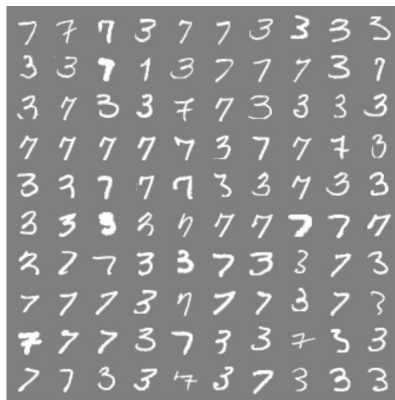
- This is a logistic regression example from Modified National Institute of Standards and Technology dataset (MNIST).
- A logistic regression model separating handwritten 3s and 7s has a 1.6% error rate on clean images.
- After applying FGSM with $\epsilon = 0.25$, the error rate rises to 99%.



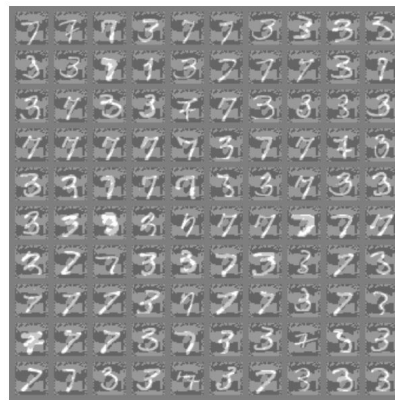
(a)



(b)



(c)



(d)

Why This Happen to Image Recognition Models?

- **Quick refresher:** the older explanation blamed nonlinearity and overfitting; Goodfellow et al. instead trace the vulnerability to linear behavior in high-dimensional spaces (Goodfellow et al., 1). The mechanism below shows exactly how that plays out mathematically:
 - Consider a linear model computing a weighted sum $w^T x$.
 - Adding a perturbation η changes this sum by $w^T \eta$.
 - If η is set to align with the sign of each weight ($\eta = \text{sign}(w)$), every one of the n input dimensions contributes a small amount to the output change.
 - With enough dimensions, many small, individually imperceptible changes add up to one large change in the model's output (Goodfellow et al., 3).
- **Precision limit connection:** digital inputs, such as 8-bit images, already discard information below a certain precision. A model should reasonably ignore changes smaller than this precision, but linear models do not, because their response scales directly with each input dimension (Goodfellow, et al., 3).
- **This is inheritable, where if model A gets tricked, then model B will likely be tricked as well:** since many different models trained on the same task tend to learn similar underlying linear decision boundaries, so a perturbation that fools one model tends to inherently fool others as well (Goodfellow et al., 8).

How Can We Defend From This?

Adversarial training: train the model on a mix of clean and adversarially perturbed examples rather than clean examples alone, by combining both training costs into a single objective so the model learns from both at once (Goodfellow et al., 5).

Modified objective: $\tilde{J}(\theta, x, y) = \alpha \cdot J(\theta, x, y) + (1 - \alpha) \cdot J(\theta, x + \eta)$

where α (alpha) is a weighting term between 0 and 1 controlling the balance between the two costs. The paper uses $\alpha = 0.5$, giving equal weight to both.

Result on MNIST: adversarial training reduced a maxout network's clean test error from 0.94% to 0.84%, and reduced its error rate on adversarial examples from 89.4% to 17.9% (Goodfellow et al., 5).

Model capacity matters: purely linear models cannot be trained to resist adversarial perturbation no matter the technique, but models with at least one hidden layer, which can represent nonlinear functions, have the capacity to learn resistance (Goodfellow et al., 4).

RBF networks as an extreme case: radial basis function (RBF) networks are naturally more resistant since they only respond confidently near training points, though this comes at the cost of weaker generalization overall (Goodfellow et al., 7).

An early hint at stopping criteria: the authors already found it useful to base stopping decisions on adversarial validation error rather than clean validation error, an idea that foreshadows the robust overfitting problem covered later in this presentation (Goodfellow et al., 5).

New Problem: Robust Overfitting

Robust overfitting, identified by Rice et al., is a failure mode specific to adversarially trained deep networks (Rice et al., 1).

Definition: after a certain point in training, robust training error keeps decreasing, but robust test error stops improving and begins to increase (Rice et al., 1).

Contrast with standard training:

- In ordinary (non-adversarial) deep learning, models trained for a long time on large architectures rarely show this kind of test-performance decline; test performance typically keeps improving or holds steady (Rice, Wong, and Kolter 1).
- In adversarial training, continuing to train past a certain point actively worsens the model's ability to resist attacks on new data, even while it appears to keep improving on the training set.

Interpretation: the model increasingly memorizes perturbation patterns specific to the training set rather than learning a decision boundary that generalizes to new adversarial inputs.

Scope: this pattern was confirmed across multiple datasets (Street View House Numbers (SVHN), Canadian Institute For Advanced Research (CIFAR)-10, CIFAR-100, ImageNet) and multiple perturbation types (L^∞ and L_2), indicating it is a general property of adversarial training rather than a dataset-specific quirk (Rice et al., 3).

Statistics on Robust Overfitting

CIFAR-10 example (Rice et al., 1, 4): a model trained with projected gradient descent (PGD)-based adversarial training reaches its best robust test error of 43.2% shortly after the first learning rate decay, then degrades to 51.4% robust test error by the end of training. This is a difference 8.2 percentage points.

To put this in perspective: nothing about the model, data, or attack changed during this gap. The only difference is additional training time, yet the model became meaningfully easier to fool, which is the opposite of what more training normally does.

Robust test error gaps by dataset, L^∞ perturbations (Rice et al., 4):

- **SVHN:** 39.0% best vs. 45.6% final (**6.6-point gap**)
- **CIFAR-10:** 43.2% best vs. 51.4% final (**8.2-point gap**)
- **CIFAR-100:** 71.9% best vs. 78.6% final (**6.7-point gap**)
- **ImageNet:** 62.7% best vs. 85.5% final (**22.8-point gap**)

TRADES example (a widely cited adversarial training algorithm): the checkpoint with the best test performance reaches 44.1% robust test error, but once training fully converges, this rises to 50.6% (Rice, Wong, and Kolter 5).

Takeaway: the gap between best and final performance is large and consistent enough that reporting only the final model, as is standard practice in non-robust deep learning, can significantly understate a model's true vulnerability.

Solution to Robust Overfitting: Early Stopping

- **Yes, there are solutions:** Rice et al. test several classical and modern techniques for combating robust overfitting on CIFAR-10 using a PreActResNet-18, including L1/L2 regularization, data augmentation, and semi-supervised learning (Rice et al., 7). Of these, early stopping stood out as the most effective and by far the easiest to implement, so this presentation focuses on it specifically.
- **What is early stopping?** A training technique where a model's training is halted, and an earlier checkpoint is selected, based on performance on a held-out validation set, rather than continuing until training loss stops decreasing (Rice et al., 5). In other words, instead of always saving the final model at the end of training, save whichever checkpoint performed best on validation data along the way, since that one is often the most robust.
- **How early stopping was implemented:**
 - First, set aside 1,000 examples from the CIFAR-10 training set to use only as a validation set, kept separate from both training and the final test set.
 - During training, after every epoch, measure the model's robust error on this validation set, meaning its error rate when the validation images are attacked by an adversary.
 - Keep a record of which epoch produced the lowest robust validation error seen so far.
 - Once training is complete, discard the final model and instead use the checkpoint saved at that lowest-validation-error epoch as the model going forward (Rice et al., 6).

Statistics and Effectiveness of Early Stopping (Rice et al., 4)

Early stopping of standard PGD adversarial training reaches 42.3% robust error on CIFAR-10 (ℓ_∞ , $\epsilon = 8/255$), actually slightly beating the reported 43.4% from the more complex TRADES algorithm.

Prevents collapse if not stopping early: Continuing to train pre-trained ImageNet models past the early stopping point destroys their performance. Robust error jumps from 62.7% $\rightarrow$ 85.5% (L_∞) and 63.0% $\rightarrow$ 94.8% (L_2).

Works without test-set leakage: Using a true hold-out validation set for early stopping achieves 46.9% robust error, essentially matching the "oracle" performance of 46.7% achieved by picking the absolute best checkpoint using the test set itself.

DATASET	NORM	RADIUS	ROBUST TEST ERROR (%)		
			FINAL	BEST	DIFF
SVHN	ℓ_∞	8/255	45.6 $\pm$ 0.40	39.0	6.6
	ℓ_2	128/255	26.4 $\pm$ 0.27	25.2	1.2
CIFAR-10	ℓ_∞	8/255	51.4 $\pm$ 0.41	43.2	8.2
	ℓ_2	128/255	31.1 $\pm$ 0.46	28.4	2.7
CIFAR-100	ℓ_∞	8/255	78.6 $\pm$ 0.39	71.9	6.7
	ℓ_2	128/255	62.5 $\pm$ 0.09	56.8	5.7
IMAGENET	ℓ_∞	4/255	85.5 $\pm$ 8.87	62.7	22.8
	ℓ_2	76/255	94.8 $\pm$ 1.16	63.0	31.8

From: Rice et al., 4

Our Research: Motivation and Scope

Open question: robust overfitting was documented by Rice et al., but why it happens at the level of what the model is actually doing internally was never explained. Our hypothesis is that it happens because the model memorizes specific perturbation patterns from the training set instead of learning a decision boundary that generalizes to new adversarial attacks. Since the perturbations generated during adversarial training depend on the model's current weights, they act as a moving target the model can gradually overfit to, rather than a stable signal it can learn general robustness from.

Why this matters: pinpointing when memorization overtakes genuine robustness learning would give a diagnostic foundation for building safeguards against robust overfitting, rather than relying on early stopping as a blunt, after-the-fact fix.

Scope of this project: this is a multi-year research direction, and testing the memorization hypothesis directly is future work, not something this summer resolves. Our concrete goal for this 8-week project is narrower: reproduce Rice, Wong, and Kolter's robust overfitting results on CIFAR-10 with PreActResNet-18 under PGD-based adversarial training, and precisely track training and test robust accuracy across epochs to locate exactly when and how the overfitting gap emerges.

What comes next in this presentation: the following slides show our progress toward that reproduction goal, which is the empirical foundation the memorization hypothesis will eventually be tested against.

(Reserved for our research findings)

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(Reserved for our research findings)

[Text goes here]

(Reserved for our research findings)

[Text goes here]

Summary and Key Takeaways

1. What is adversarial, why it is a problem, and what caused it: an adversarial example is an input with a tiny, invisible change that makes a model confidently give the wrong answer. This matters because it means a highly accurate model can still be reliably fooled. The cause is not complexity, but linearity: networks respond so predictably to input changes that many tiny changes can add up into one large, damaging shift (Goodfellow et al., 1).
2. How can we defend from adversarial attacks? Adversarial training: train the model on a mix of clean and adversarially perturbed examples instead of clean examples alone, so it learns to resist the same kind of attack it faces. This measurably reduced both clean and adversarial error rates (Goodfellow et al., 5).
3. During adversarial training, we have a new problem: robust overfitting: past a certain point, the model keeps improving on training data but gets worse on new, unseen data, the opposite of what normally happens in standard training (Rice et al., 1).
4. So the solution is early stopping: what is it, and how effective is it: stop training early and keep the checkpoint that performed best on a validation set, rather than the final model. This simple approach nearly matched the best possible test result and outperformed several more complex techniques, with no extra data or architecture changes needed (Rice et al., 6-7).
5. [Reserved for our research findings]
6. [Reserved for our research findings]

References

Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and Harnessing Adversarial Examples." International Conference on Learning Representations (ICLR), 2015.

Rice, Leslie, Eric Wong, and J. Zico Kolter. "Overfitting in Adversarially Robust Deep Learning." Proceedings of the 37th International Conference on Machine Learning, PMLR, vol. 119, 2020.

Veerabadran, Vijay, et al. "Subtle Adversarial Image Manipulations Influence Both Human and Machine Perception." Nature Communications, vol. 14, no. 1, 2023, article 4933. Springer Nature, 15 Aug. 2023, <https://doi.org/10.1038/s41467-023-40499-0>

Thank you!

Any Questions?

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